

</> CODESTAXH

COMP4768 Group Project, Prof. Pranjal Patra

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Table of Contents:

1. Description
 2. What makes CodeStaxh different
 3. Features
 4. Tech stack
 5. Installation guide
 6. How to use CodeStaxh
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Description

Codestaxh is a cross platform code snippet library designed for developers who need quick access to reusable code patterns. The app solves the "I know I wrote this before, but where?" problem by providing a centralized, searchable library of code snippets accessible from both web and mobile devices.

Web Version: Optimized for creating and editing snippets.

Mobile Version: Designed for quick reference: browse, search, view, and copy snippets on the go with offline access.

The app enables team collaboration, allowing developers to share snippets in real time, upvote the best solutions, and build an organized knowledge base specific to their projects and workflows.

What makes it different?

- + actually works offline
- + syncs in real-time
- + team collaboration built-in

- + AI that's not mid
- + browser extension for Stack Overflow and Github
- + UI that goes hard
- + cross-platform (web + mobile) **basically**: you write code once, access it everywhere. no more "wait where did i save that regex" moments.

Features Syntax Highlighting

15+ languages with proper highlighting. your code looks as good as it runs.

Smart Search

Find anything by title, tags, language, or even code content.

Real-time Sync

Edit on web, update yourself and your team immediately.

Team Collab

Share snippets with your team. no more dm'ing code through slack.

AI-Powered

Automatically tag your snippets or get quick and concise descriptions of snippets.

Browser Extension

Save snippets directly from Stack Overflow. One click and you're done.

Tech Stack Frontend

- Firebase Auth - Google/GitHub/Email sign-in
- Flutter – smooth and UX operable on nearly all devices/
- Material 3 – Three-dimensional appearance
- Custom Animations – enhance user experience during loading wait times.

Backend

- Firebase Auth - Google/GitHub/Email sign-in

- Firestore - real-time sync
- Hive - offline storage
- Riverpod - state management
- Gemini API – ai features (descriptions, auto-tagging)

Platforms

- Web (deployed on Firebase Hosting)
- Mobile (Android)
- Chrome Extension

Installation

Web: just go to gr11-codestaxh-fall-2025.web.app ← it's literally that easy

Mobile: simply install CodeStaxh.apk in this repository's root directory.

How to use

1. **Sign in** - google, github, or email.
2. **Create snippet (webapp)** - paste code, add tags, done.
3. **Search** – Find snippets from yourself, others, or your team.
4. **Share** - create a team, invite people, share snippets.

architecture

lib/

```
└─ models/      # data models
└─ controllers/ # business logic
└─ views/       # UI screens
| └─ web/       # web-specific views
| └─ mobile/    # mobile-specific views
```

| └─ shared/ # shared views
└─ widgets/ # reusable components
└─ theme/

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